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Trace the following operations in Circular Double Linked List

i) Insert 46, 93, 76, 35 in the Beginning

ii) Insert after 76

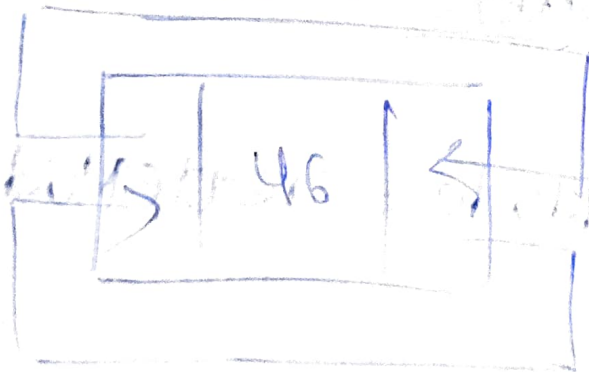
Insert at the end 100

Insert at the end 200

iii) Delete 100, 400, 76

i) Insert 46 at the beginning

head/tail

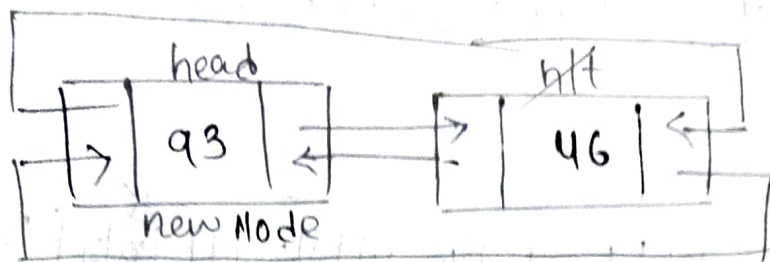


if (head == NULL)

head = new Node;

tail = new Node;

Insert 93:

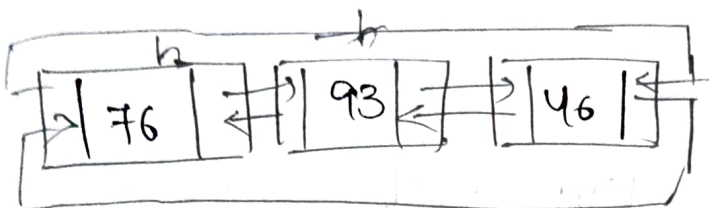


```

head → next = head;
head → prev = head;
}

```

Insert 76:



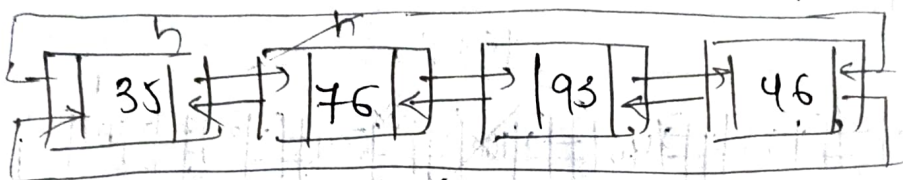
$\text{newNode} \rightarrow \text{next} = \text{head};$

$\text{newNode} \rightarrow \text{prev} = \text{tail};$

$\text{head} \rightarrow \text{prev} = \text{newNode};$

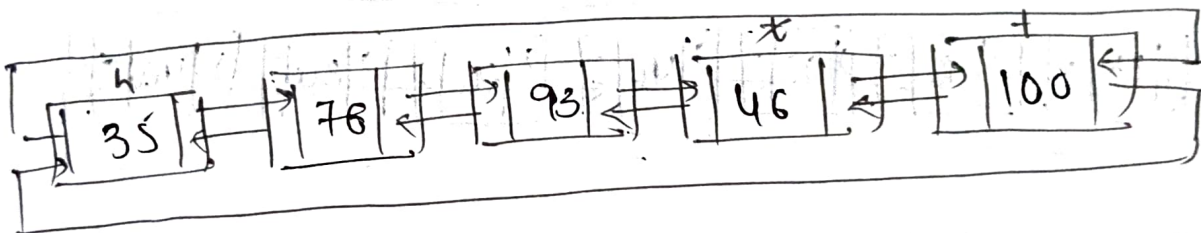
$\text{tail} \rightarrow \text{next} = \text{newNode};$

Insert 35:



$\text{head} = \text{newNode};$

ii) Insert at the end:



$\text{newNode} \rightarrow \text{prev} = \text{tail};$

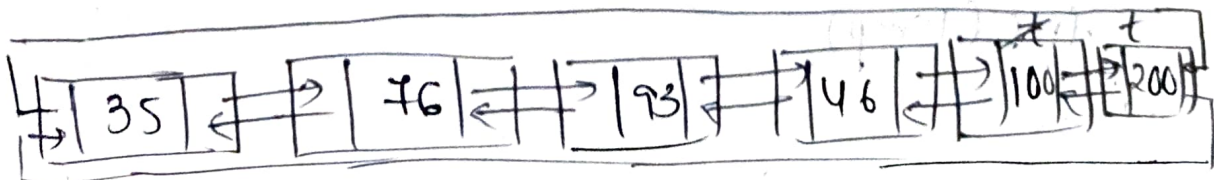
$\text{newNode} \rightarrow \text{next} = \text{head};$

$\text{tail} \rightarrow \text{next} = \text{newNode};$

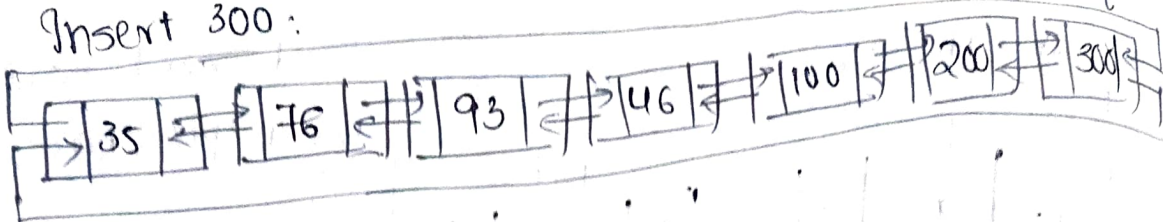
$\text{head} \rightarrow \text{prev} = \text{newNode};$

$\text{tail} = \text{newNode};$

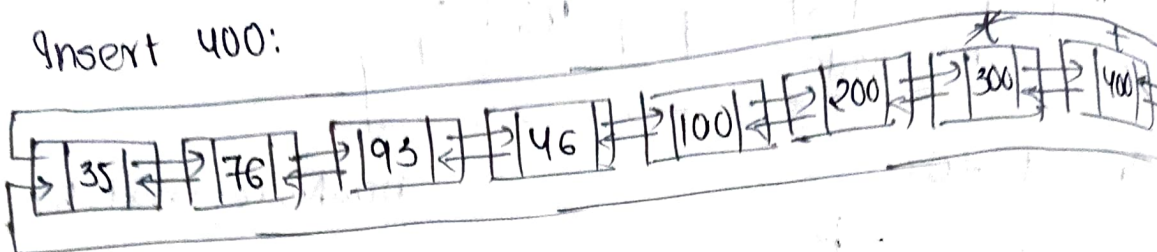
Insert 200:



Insert 300:



Insert 400:

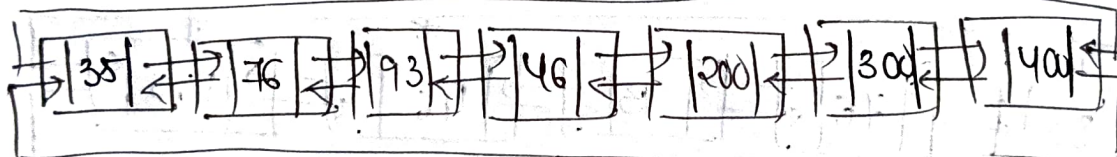
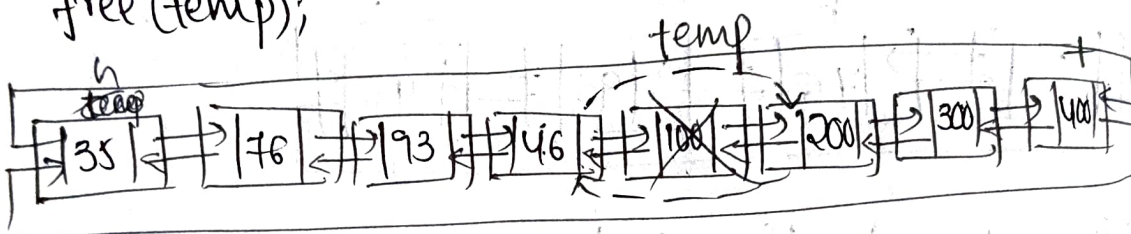


ii) Delete 100: $\emptyset$

$\text{temp} \rightarrow \text{prev} \rightarrow \text{next} = \text{temp} \rightarrow \text{next};$

$\text{temp} \rightarrow \text{next} \rightarrow \text{prev} = \text{temp} \rightarrow \text{prev};$

$\text{free}(\text{temp});$



Delete 400; at the end.

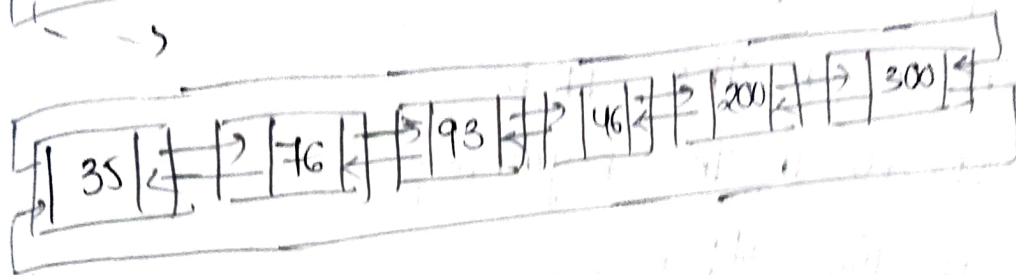
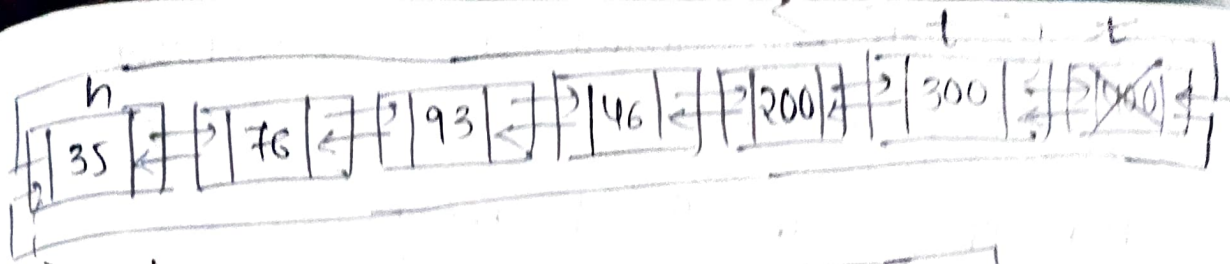
$\text{tail} = \text{tail} \rightarrow \text{prev};$

$\text{tail} \rightarrow \text{next} = \text{head};$

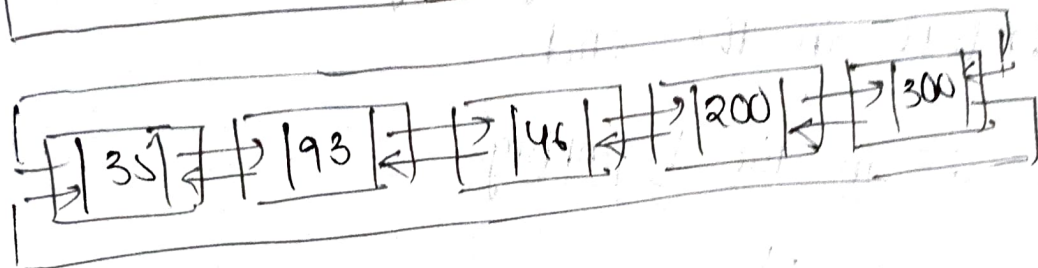
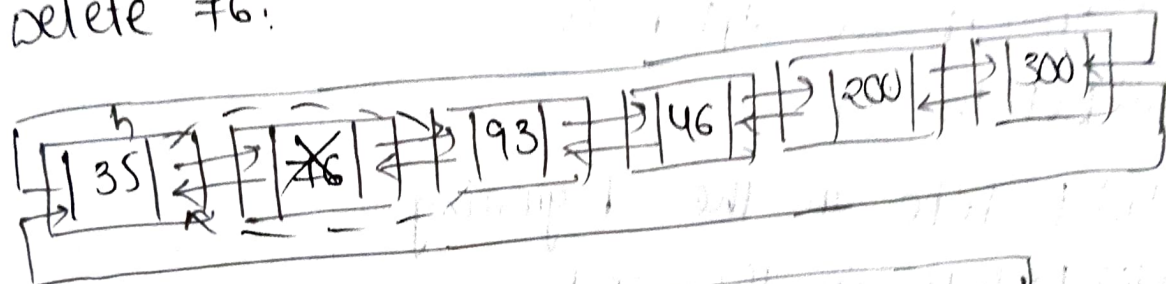
$\text{head} \rightarrow \text{prev} = \text{tail};$

}

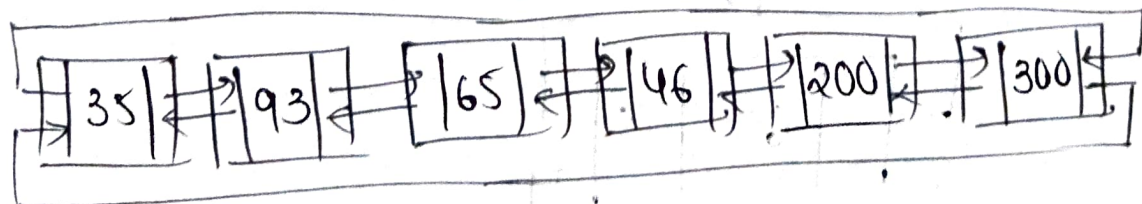
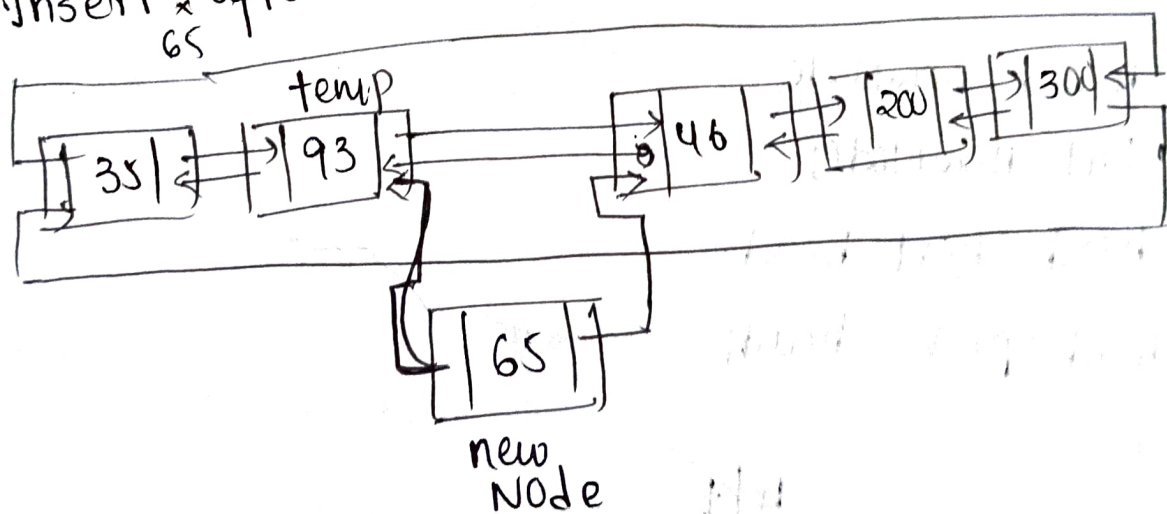
$\text{free}(\text{temp});$



Delete 76:



Insert 65 after 93:

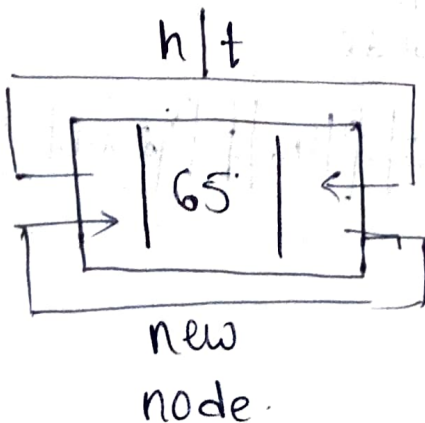


Q. Trace the following instructions for circular Double Linked List.

- i) Insert 65 in the beginning.
- ii) Insert 43 in the End.
- iii) Insert 76 in the beginning.
- iv) Insert 100 after 65
- v) Insert 200 after 100
- vi) Delete 43
- vii) Delete in the beginning
- viii) Delete in the end.

i) Insert 65 in the beginning:

```
if (head == NULL)
{
    head = newNode;
    tail = newNode;
    head → next = head;
    head → prev = head;
}
```



ii) Insert 43 in the End:

if (head == NULL)

{ -- }

else {

newNode → prev = tail

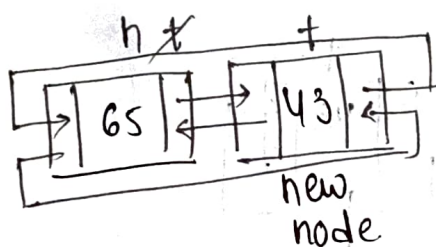
newNode → next = head;

tail → next = newNode;

head → prev = newNode;

tail = newNode;

}



iii) Insert 76 in the beginning:

if (head == NULL)

{ -- -- }

else {

newNode → next = head;

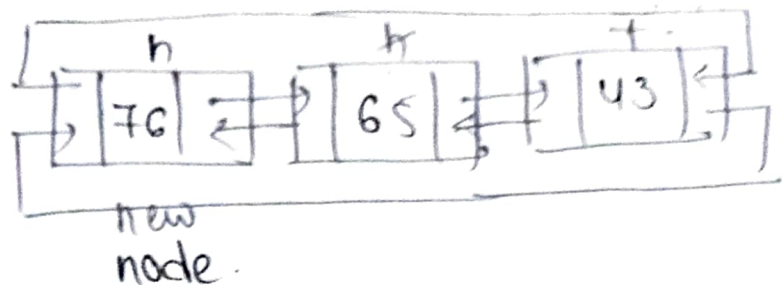
newNode → prev = tail;

head → prev = newNode;

tail → next = newNode;

head = newNode;

}



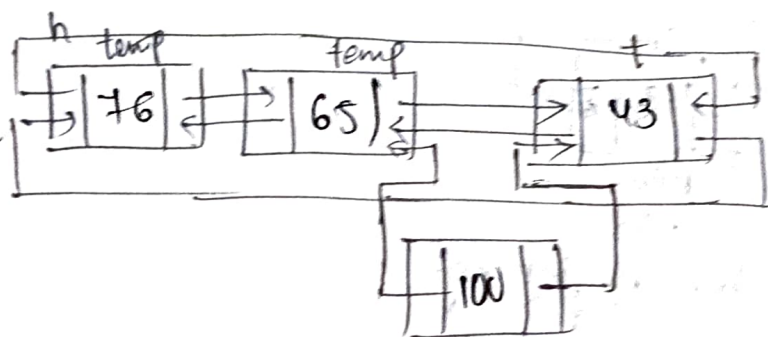
iv) Insert 100 after 65:

$\text{newNode} \rightarrow \text{prev} = \text{temp};$

$\text{newNode} \rightarrow \text{next} = \text{temp} \rightarrow \text{next};$

$\text{temp} \rightarrow \text{next} \rightarrow \text{prev} = \text{newNode};$

$\text{temp} \rightarrow \text{next} = \text{newNode};$



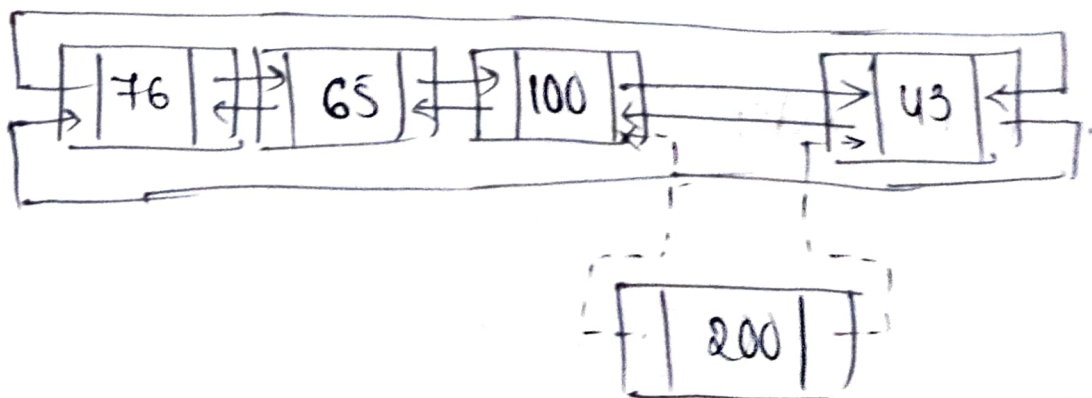
v) Insert 200 after 100:

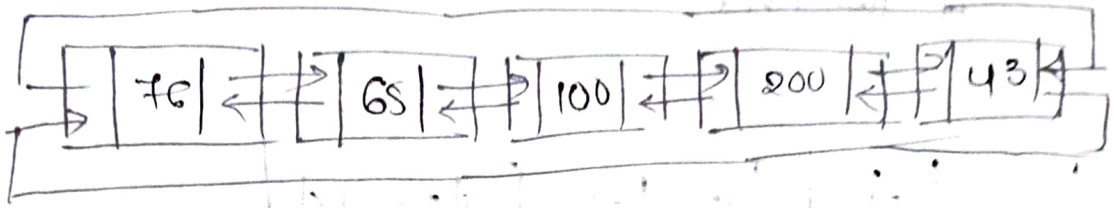
$\text{newNode} \rightarrow \text{prev} = \text{temp};$

$\text{newNode} \rightarrow \text{next} = \text{temp} \rightarrow \text{next};$

$\text{temp} \rightarrow \text{next} \rightarrow \text{prev} = \text{newNode};$

$\text{temp} \rightarrow \text{next} = \text{newNode};$





vi) Delete 43:

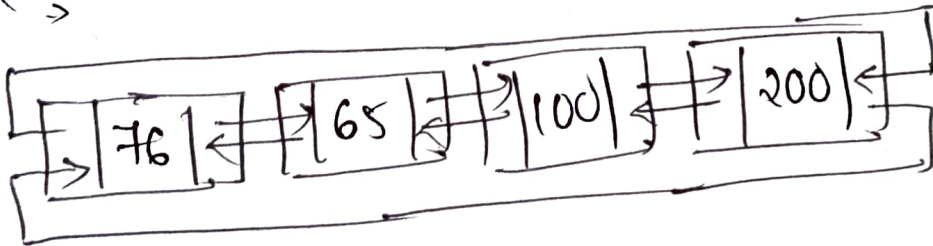
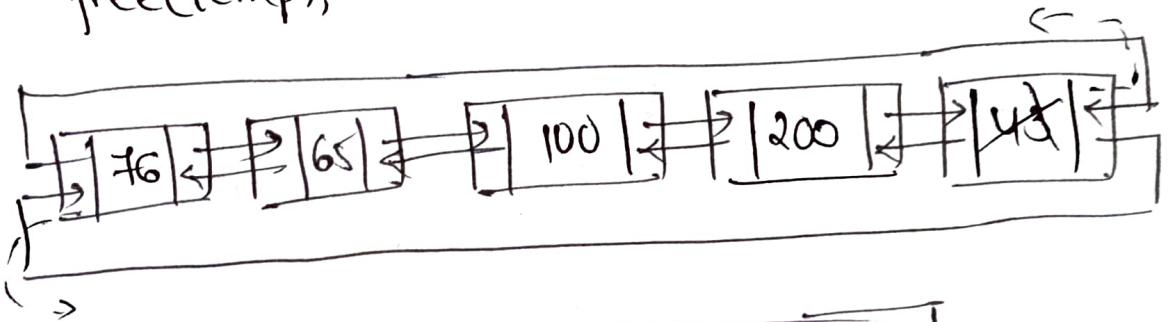
tail = tail → prev;

tail → next = head;

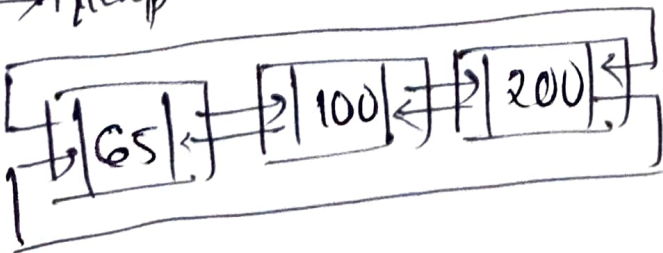
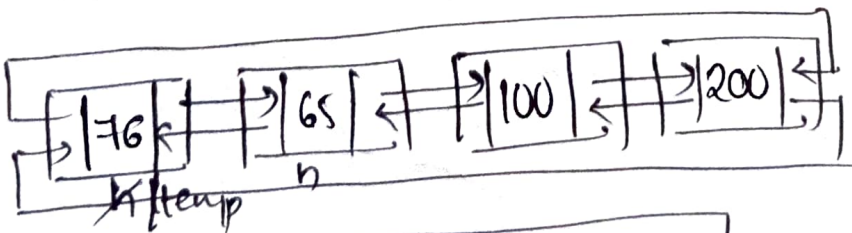
head → prev = tail;

}

free(temp);



vii) Delete in the beginning:



vin) Delete at the End

